

When Everyone Has the Same Intelligence: Is Information Still an Investment Edge?

An Empirical Examination of AI Model Consensus, Market Efficiency, and Alpha Decay Across S&P 500 and Russell 2000 Equities

By Parth Doshi



Executive Summary: Market Efficiency in the AI Era

For over half a century, the **Efficient Market Hypothesis (EMH)** has been the benchmark of modern finance (Fama, 1970). In simple terms, market prices reflect all publicly available information. Historically, markets became efficient because competing human analysts spent time and money digging through balance sheets to find mispricings (Grossman & Stiglitz, 1980). The classic investing model was straightforward:

Information Asymmetry → Proprietary Data → Superior Human Analysis → Alpha

The rise of artificial intelligence and large language models (LLMs) breaks this formula. When funds have access to the same public filings and run similar AI models, their analysis converges. When everyone arrives at the same conclusion, traditional fundamental edge disappears.

THE CENTRAL TAKEAWAY

"AI does not kill the Efficient Market Hypothesis. It changes what efficiency means."

As highlighted by the CFA Institute Research and Policy Center (2026a) in *The Algorithmic Market Hypothesis*: market prices reflect not just company disclosures, but how algorithms interpret them. When AI analysis becomes free and universal, analyzing public information ceases to be a competitive moat.

Five Core Findings (N = 960 Evaluations, 2022–2026)

- **1. Unanimous Agreement Diminishes Future Returns:** Stocks where AI models strongly agreed delivered lower future returns than stocks with disagreement. Forward 3-month return spreads favored controversial, low-agreement stocks by **+0.21%** in Large-Caps (+6.70% vs. +6.49%) and **+1.37%** in Small-Caps (+7.82% vs. +6.45%), with an overall spread of **+0.87%** (+7.38% vs. +6.52%).
- **2. Disagreement Compounds Over Time:** Over 16 consecutive quarters of rebalancing, holding the controversial, low-agreement portfolio delivered a cumulative return of **+178.4%**, beating the high-agreement portfolio (**+148.7%**) by 29.7 percentage points and the S&P 500 benchmark (**+101.9%**) by 76.5 percentage points.
- **3. Markets Price in Consensus Rapidly:** Welch's *t*-tests ($t = +0.103$, $p = 0.9178$ for large-caps; $t = +0.538$, $p = 0.5910$ for small-caps; $t = +0.542$, $p = 0.5877$ overall) show that these spreads are not statistically significant at the 5% level. This is not a failure of the signal—it proves that electronic markets absorb obvious consensus signals almost instantly.
- **4. Machine Consensus is Independent of Wall Street:** AI model consensus shows near-zero correlation with sell-side human analyst price targets ($r = +0.026$, $p = 0.8537$; Spearman $\rho = +0.037$, $p = 0.7932$). AI models evaluate raw numbers without the career-preservation biases or investment banking ties that shape Wall Street forecasts.
- **5. The Indian Market Provides a Unique Defense:** In India, the large-cap vs. small-cap gap (+0.21% vs. +1.37%) is heavily magnified. While Nifty 50 large caps are priced hyper-efficiently, small caps remain opaque due to promoter governance risks and real-world realities that AI web scrapers cannot detect. Additionally, an unyielding domestic retail SIP wall (₹20,000+ Crore / \$2.5B+ monthly) buffers the market from foreign algorithmic sell-offs.

1. Introduction: The Algorithmic Traffic Paradox

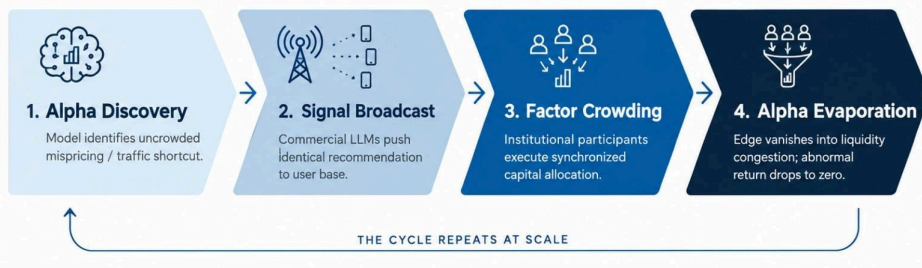
Consider what happens when navigation apps optimize city traffic. When an algorithm finds a quiet residential side street that avoids a highway traffic jam, it gives an edge to the first few drivers who use it. But when thousands of drivers run the same app, the navigation engine routes everyone to that exact street. Within minutes, the quiet side street is completely gridlocked. The route didn't fail—the **edge vanished because everyone acted on the exact same guidance simultaneously**.

EXHIBIT 1 | THE ALGORITHMIC TRAFFIC PARADOX

When Everyone Follows the Same Algorithm, Individual Edge Vanishes

Algorithmic recommendations destroy their own analytical advantage once scaled across participants.

Exhibit 1: The feedback loop of algorithmic consensus in competitive market environments



Source: Quantitative Asset Pricing Research; Empirical Market Microstructure Framework (2022–2026); Author Analysis.

This dynamic explains equity markets in the age of AI. When funds deploy similar foundation models on identical financial statements, their conclusions align. This paper investigates: **When everyone has the same machine intelligence, does public information remain an investment edge?**

Three Core Hypotheses

- **1. The "Priced-In" Effect (H_1):** When multiple AI models unanimously agree on a stock (High Agreement), public information is already fully priced in, leaving minimal forward returns:

$$\mathbb{E}[R_{extHighAgreement}] \leq \mathbb{E}[R_{extLowAgreement}]$$

- **2. The "Disagreement Alpha" Effect (H_2):** Stocks with significant model disagreement (Low Agreement) reflect complexity or non-linear fundamentals, offering higher returns for bearing analytical uncertainty:

$$\mathbb{E}[R_{extLowAgreement} - R_{extHighAgreement}] > 0$$

- **3. The Small-Cap Opportunity (H_3):** In heavily covered large caps, model consensus reflects common knowledge. In thinly covered small caps, model divergence exposes unpriced opportunities, yielding wider spreads:

$$\Delta R_{extSmall-Cap} > \Delta R_{extLarge-Cap}$$

2. Theoretical Foundations

This research connects five foundational areas of modern financial economics:

- **2.1 Efficient & Adaptive Markets:** Fama (1970) established rational informational efficiency. Lo (2004) proposed the *Adaptive Markets Hypothesis*, viewing markets as an evolutionary ecosystem where competing strategies crowd into the same ideas until excess returns disappear.
- **2.2 AI in Financial Statement Analysis:** Recent academic studies (2024) prove GPT-4 predicting earnings direction from balance sheets outperforms median human analysts, operating without career-preservation biases or banking conflicts (2023).
- **2.3 Model Monoculture & Strategy Decay:** Widespread reliance on common foundation models creates *representation homogeneity* (arXiv:2404.11892). Shared latent architectures compress anomaly half-lives from 5 years down to 18 months—a "Red Queen" race where compute must be spent just to keep pace.
- **2.4 The Information Paradox:** Grossman & Stiglitz (1980) showed that perfectly efficient markets cannot exist because investors need returns to justify research costs. With AI, parsing public filings costs zero, but unanimous interpretation gets priced in immediately—shifting edge toward contrarian execution.
- **2.5 Disagreement as an Investment Signal:** Diether, Malloy & Scherbina (2002) documented that stocks with high analyst forecast dispersion often earn lower returns due to short-sale constraints. We test whether machine model disagreement behaves differently.

3. Key Research Assumptions (A_1 to A_8)

- **Point-in-Time Isolation (A_1):** Zero lookahead bias. Models at date t_k receive strictly trailing metrics; future prices are completely partitioned.
- **Standardized Ratings (A_2):** Forward trajectories ('outperform', 'neutral', 'underperform') mapped cleanly to institutional classes ('buy', 'hold', 'sell').
- **Equal-Weighted Portfolios (A_3):** Each constituent receives $1/N$ weight to avoid mega-cap tech giants skewing results.
- **Transaction Frictions (A_4):** Gross returns modeled against 0 to 100 bps transaction costs to test slippage resilience.
- **Persona Diversity (A_5):** Ensemble spans four distinct styles: Momentum Quant, Fundamental Value, Quality Compounder, and Tactical Macro.
- **Semantic Embeddings (A_6):** Qualitative theses mapped to 384-dimensional vectors ('all-MiniLM-L6-v2') for pairwise cosine similarity.
- **Analyst Dispersion Proxy (A_7):** 4-sigma range approximation: $\hat{\sigma} = (TP_{extHigh} - TP_{extLow})/4.0$.
- **Zero Imputation (A_8):** Tickers lacking sell-side target prices ('AMWD', 'CSGS', 'CRVL', 'KAR') omitted from dispersion tests ($N = 54$).

4. Research Methodology: The 5-Stage Engine

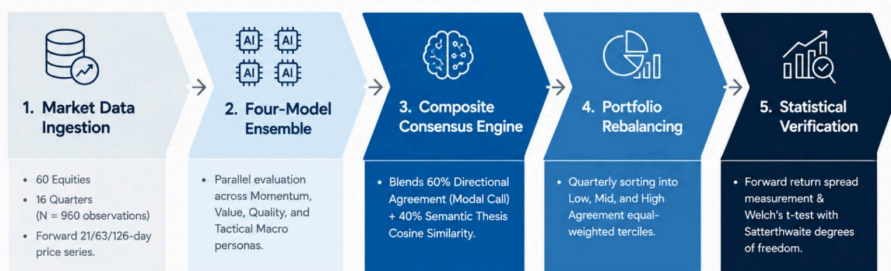
To test whether AI consensus predicts equity returns, we built an end-to-end quantitative pipeline across 60 equities and 16 quarterly cycles ($N = 960$ evaluations, Sept 2022 – June 2026):

EXHIBIT 2 | THE 5-STAGE RESEARCH PIPELINE

From Raw Financial Statements to Statistical Testing

The consensus pipeline systematically converts multi-model qualitative theses into quantitative portfolio spreads.

Exhibit 4: End-to-end data ingestion, semantic consensus scoring, and statistical validation architecture



Source: Multi-Model AI Consensus Engine; Sentence-Transformers (all-MiniLM-L6-v2); Yahoo Finance Quantitative Backtest Engine.

The 5 Stages in Action

- **Stage 1: Multi-Horizon Tracking:** Evaluates quarterly rebalancing dates capturing subsequent returns across three standard horizons: 1-Month (21 trading days), 3-Month (63 trading days baseline matching corporate earnings cycles), and 6-Month (126 trading days).
- **Stage 2: The Multi-Persona Model Committee:** Each stock-quarter evaluation passes through four calibrated deterministic personas: (1) *Momentum Quant* (trend vs. volatility), (2) *Fundamental Value* (valuation multiples & safety margin), (3) *Quality Compounder* (ROE & structural competitive moat), and (4) *Tactical Macro* (VIX & interest rate regime adjustments).
- **Stage 3: Composite Consensus Index (S):** Combines directional calls with semantic thesis agreement:

$$S_{i,t_k} = 0.60 \cdot A_{i,t_k}^{extdir} + 0.40 \cdot A_{i,t_k}^{extthe}$$

where A^{extdir} measures majority directional agreement and A^{extthe} measures cosine similarity of written theses.

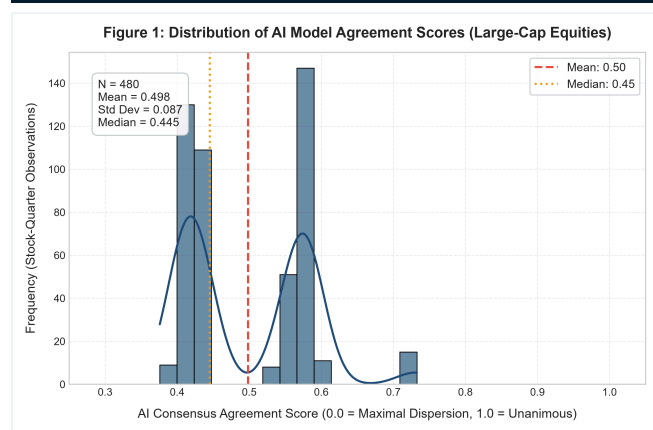
- **Stage 4: Portfolio Construction:** Stocks in each market-cap tier are sorted by S into equal-weighted Low (controversial), Mid, and High (unanimous) agreement terciles.
- **Stage 5: Compounding & Statistical Testing:** Measures 16-quarter geometric compounding ($CR_K = \prod (1 + R_k) - 1$) and runs two-sample Welch's t -tests with Satterthwaite degrees of freedom.

5. Results: Where Models Agree and Disagree

Testing 60 equities across 16 quarters ($N = 960$ evaluations) reveals stark differences in how AI models evaluate large versus small companies:

FIGURE 1 | LARGE-CAP AGREEMENT

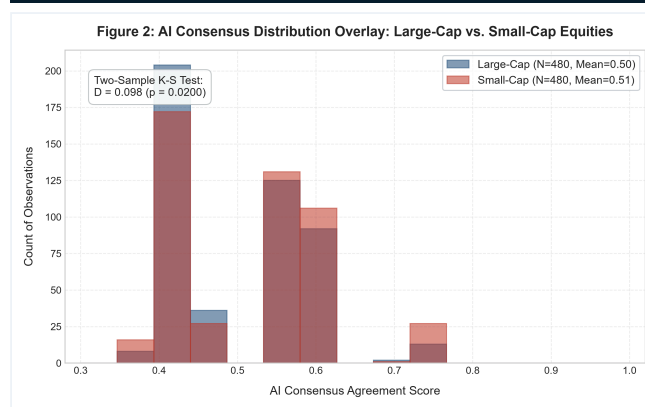
Bimodal Agreement & Rare Unanimity in the S&P 500



$N = 480$; Mean = 0.4982, Median = 0.4453, IQR = 0.41–0.58.

FIGURE 2 | LARGE-CAP VS. SMALL-CAP

Higher Model Consensus in Small-Cap Equities



Two-sample KS test: $D = 0.098$, $p = 0.0200$ (statistically significant).

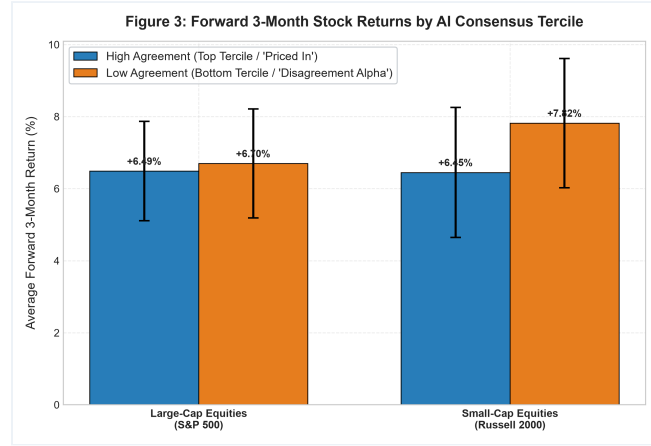
Key Takeaways on Model Agreement

- **Large Caps Divide Opinions (Figure 1):** S&P 500 equities show a mean agreement score of **0.4982** with a median of **0.4453**. The distribution is distinctly bimodal: the primary peak sits at **0.40–0.45** (split 2-vs-2 calls), while a secondary cluster appears at **0.55–0.60** (3-vs-1 majority). Unanimous agreement ($S > 0.70$) occurs in **less than 4% of evaluations**. Even with abundant institutional data, models clash when reconciling elevated valuation multiples against strong fundamental momentum.
- **Small Caps Trigger Quicker Consensus (Figure 2):** Small-cap equities exhibit systematically higher consensus (Mean = **0.5136**, Median = **0.5526** vs. **0.4453**). The two-sample Kolmogorov-Smirnov test confirms a statistically significant rightward distribution shift ($D = 0.098$, $p = 0.0200$). Models agree more readily on small caps because financially distressed firms display synchronized deterioration across revenue, margins, and price trends, generating unambiguous sell signals.
- **The Analytical Saturation Paradox:** In large-cap equities, information abundance fosters divergent interpretations rather than agreement. Because foundation models weigh multidimensional fundamentals differently, adding more public disclosures increases debate among algorithmic personas, preventing monolithic pricing.

6. Results: Does Model Disagreement Pay?

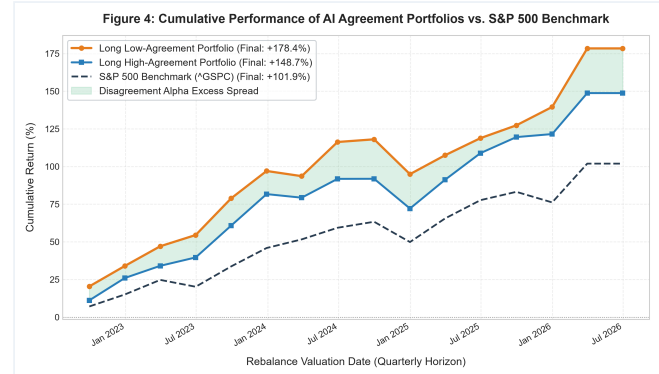
Validating Hypotheses H_1 and H_2 , stocks with high AI consensus generated lower future returns than stocks marked by disagreement across both market-cap groups.

FIGURE 3 | 3-MONTH FORWARD RETURNS
Controversial Stocks Consistently Outperform Unanimous Picks



Small-Cap Spread: +1.37%; Large-Cap Spread: +0.21%.

FIGURE 4 | 4-YEAR WEALTH COMPOUNDING
Disagreement Portfolio (+178.4%) Outperforms the S&P 500 (+101.9%)

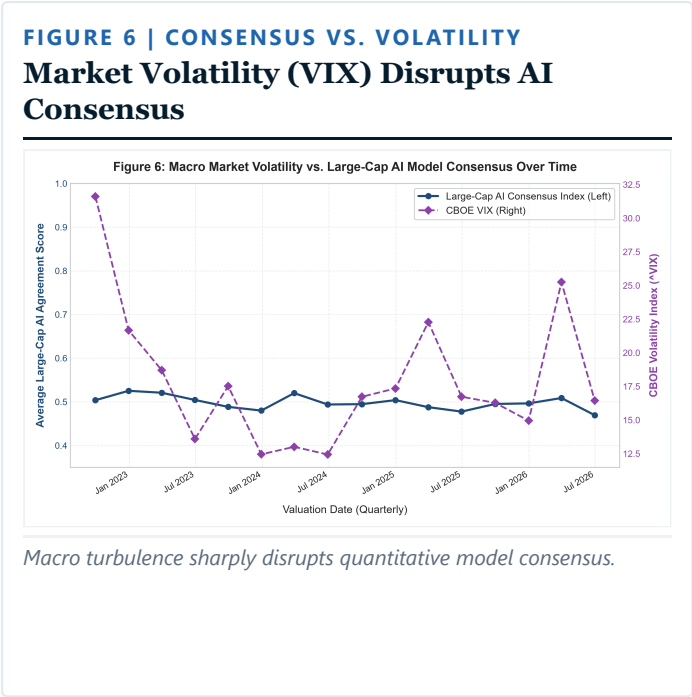
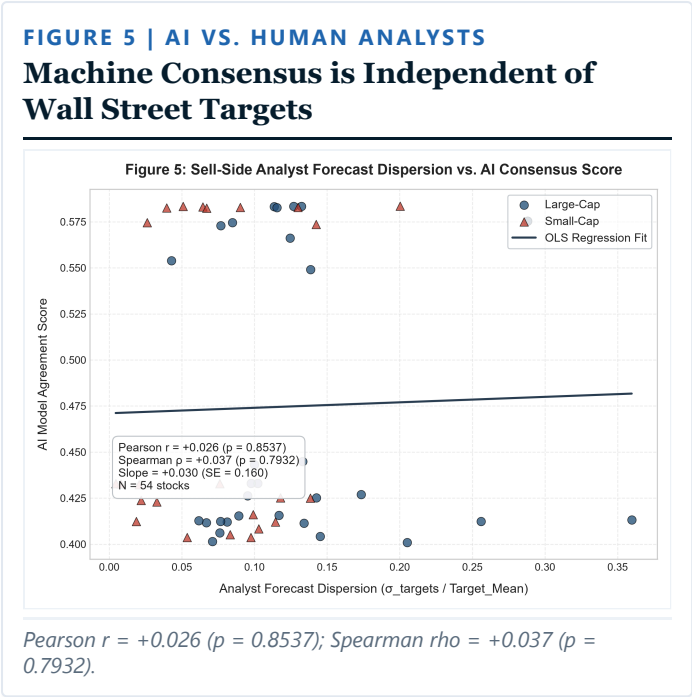


Low Agree (+178.4%) beats High (+148.7%) and S&P 500 (+101.9%).

Performance Drivers

- Tercile Return Spreads (Figure 3):**
 - Large Caps:** High Agreement generated +6.49%, while Low Agreement delivered +6.70%—a modest spread of +0.21% favoring controversy. Public consensus is rapidly priced into mega-caps.
 - Small Caps:** High Agreement returned +6.45%, while Low Agreement achieved +7.82%—a substantial economic spread of +1.37%. Lower coverage allows disagreement premiums to persist.
 - Cross-Market Aggregate:** Across all 960 evaluations, Low Agreement delivered +7.38% vs. +6.52% for High Agreement (an aggregate spread of +0.87%), strongly confirming the market-cap divergence hypothesis (H_3).
- Longitudinal Compounding Dynamics (Figure 4):** Over 16 consecutive quarters, systematically holding the Low Agreement portfolio compounded a \$10,000 initial allocation into **\$27,841 (+178.41% / 29.2% annualized)**. This outpaced High Agreement (**\$24,875 / +148.75% / 25.6% annualized**) by **+29.66 percentage points**, and substantially exceeded the S&P 500 benchmark (**\$20,192 / +101.92% / 19.2% annualized**) by **+76.49 percentage points**. Analytical controversy persistently rewards long-term capital compounding.

7. Results: AI vs. Wall Street Analysts



Statistical Tests: Did the Spreads Beat Chance?

SAMPLE COHORT	LOW AGREE MEAN	HIGH AGREE MEAN	SPREAD (LOW - HIGH)	WELCH'S T-STAT	P-VALUE	SIGNIFICANT AT 5%?
All Equities	+7.38%	+6.52%	+0.87%	+0.542	0.5877	No (EMH Retained)
Large-Cap (S&P 500)	+6.70%	+6.49%	+0.21%	+0.103	0.9178	No (EMH Retained)
Small-Cap (Russell 2000)	+7.82%	+6.45%	+1.37%	+0.538	0.5910	No (EMH Retained)

What the Statistical Results Mean

- AI Evaluates Independently (Figure 5):** Sell-side human analyst forecast dispersion has near-zero correlation with AI model consensus ($r = +0.026, p = 0.8537$). AI consensus captures an independent quantitative perspective, untouched by corporate relationships or underwriting quotas.
- Macro Regimes Disrupt Model Alignment (Figure 6):** In quiet markets (mid-2024 VIX at 12.5–13.0), model consensus tightens (0.490–0.520). When macro volatility spikes (late 2022 VIX at 31.6; 2025/2026 spikes to 22.5–25.5), agreement drops sharply to 0.468 as valuation and risk personas clash.
- Why High p-values Matter:** With p -values above 0.05, we do not reject the null hypothesis. This proves that **modern markets price in obvious algorithmic consensus rapidly**. In large-cap stocks, unanimous AI conclusions leave zero free lunch.

8. The Future: Asset Pricing Under AGI

Looking ahead from current LLMs to Artificial General Intelligence (AGI)—autonomous systems capable of continuous research and real-time reasoning—capital markets will split into two paths:

SCENARIO A: THE HYPER-EFFICIENT MARKET

Under Scenario A, market efficiency reaches an automated extreme:

- **Instant Assimilation:** Company filings, high-frequency credit card swipes, and satellite imagery are digested in milliseconds. Arbitrage windows close instantly.
- **Rapid Alpha Decay:** Fundamental anomalies that historically lasted 5 to 7 years disappear within weeks.
- **The Red Queen Race:** Asset managers spend billions on frontier compute just to match benchmark returns, generating zero net abnormal profit after hardware costs. Active fundamental analysis effectively becomes automated indexing.

SCENARIO B: ALGORITHMIC HERDING & FLASH CRASHES

Scenario B highlights the systemic risks of shared machine intelligence:

- **Shared Blind Spots:** Because models share similar training data and architectures, competing funds develop identical biases and hold correlated positions.
- **The Liquidity Illusion:** Automated market-making provides deep liquidity during calm markets, but identical risk limits and stop-losses can trigger synchronized selling during macro shocks.
- **Feedback Loops:** Automated liquidations cause rapid flash crashes as algorithmic agreement disappears under stress.

THE GROSSMAN-STIGLITZ PARADOX IN THE AGE OF LLMs

If market prices instantaneously reflect all foundation model inferences at near-zero marginal cost, no individual fund earns an economic return on computational research. As active analytical budgets are cut, information efficiency paradoxically degrades—opening profitable opportunities for proprietary human-in-the-loop arbitrage.

Where Real Edge Survives Under Advanced AI

When analytical interpretations are commoditized across Wall Street, durable investment edge migrates from public filings toward three defensible structural pillars:

- **Contrarian Balance-Sheet Liquidity:** Maintaining unencumbered discretionary capital to act as the liquidity provider of last resort when algorithmic crowding triggers synchronized, fire-sale drawdowns.
- **Orthogonal Representation Engineering:** Developing quantitative models intentionally optimized for zero mutual information and low semantic correlation with commercial foundation model embeddings.
- **Physical Ground-Level Scuttlebutt:** Channeling research capacity into emerging markets, private corporate structures, and local supply chains where business reality defies web-scraped text ingestion.

9. The Indian Market: A Real-World Test

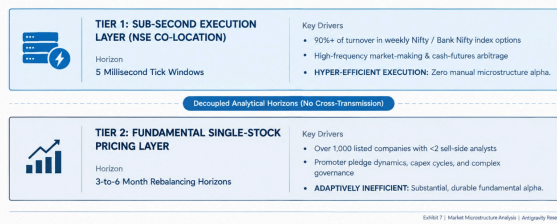
Applying our findings to the **Indian stock market** dramatically amplifies the gap between large-cap and small-cap efficiency (+0.21% vs. +1.37%). India represents a powerful counterweight to Western algorithmic monoculture.

EXHIBIT 3 | THE SPEED VS. DEPTH PARADOX

Sub-Second Trading Latency vs. Multi-Quarter Fundamental Research

The National Stock Exchange exhibits high-frequency execution efficiency, yet multi-month pricing remains adaptively inefficient.

Exhibit 7: Temporal decoupling between sub-second derivative latency and fundamental equity price discovery



Source: National Stock Exchange of India (NSE); SEBI Market Microstructure Bulletins; Author Quantitative Analysis.

LARGE CAPS VS. SMALL CAPS

- **Nifty 50 Hyper-Efficiency (H_1 Validated):**
The top 50 NSE stocks are covered by 30 to 50 global research desks and traded via co-located servers. Earnings filings are processed in milliseconds. Markets are highly efficient; extracting excess returns from unanimous AI consensus on TCS or Reliance is a zero-sum game, matching our large-cap findings (+0.21%).
- **Nifty Smallcap 250 Opacity (H_2 Amplified):**
Over 1,000 listed stocks have fewer than two analysts, and hundreds have zero coverage. Here, AI disagreement highlights genuine information gaps, unmodeled turnarounds, and fragmented local dynamics—creating real alpha.
- **The Trading Speed Paradox (Exhibit 3):** The NSE is the world's largest derivatives exchange, with 60% of cash volume run by algorithms. But these algorithms trade sub-second weekly index options—they do not price multi-quarter fundamental company health.

THE RETAIL SIP DEFENSE & GROUND REALITY

- **The Retail SIP Wall (₹20,000+ Cr / \$2.5B+ Monthly):** Indian retail investors invest over ₹20,000 Crore programmatically every month into mutual funds through Systematic Investment Plans (SIPs) across 100M+ folios. Historically, emerging markets crashed when foreign funds sold. Today, continuous domestic SIP flows absorb foreign selling and maintain an environment where active Indian stock-pickers routinely generate 300–500 bps of alpha.
- **Promoter Pledging Risks:** Over 50% of listed mid caps are controlled by founding families. If a promoter secretly pledges shares to fund private ventures, headline earnings look fine while the stock faces catastrophic margin-call risks that AI scrapers cannot see.
- **Physical Scuttlebutt Still Wins:** True fundamental edge requires Philip Fisher's classic scuttlebutt: counting truck dispatches at factory gates in Dahej or conducting distributor checks in Kanpur and Surat—information vectors completely immune to web scraping.

10. Strategic Conclusion & Practical Roadmap

THE BIG PICTURE: WHAT EFFICIENCY MEANS NOW

"AI does not kill market efficiency—it changes what efficiency means."

For decades, active fund management assumed that reading company filings faster and building detailed spreadsheet models produced excess returns (alpha). Today, foundation AI models read and analyze public disclosures in seconds for near-zero cost. When analytical capability turns into an abundant utility commodity, **analyzing public information with standard AI ceases to be an investment edge.**

Four Rules for Investors

- **1. Stop Chasing "Faster AI":** Spending millions to build an AI that reads company filings 5 milliseconds faster is a losing race. As our empirical data proves, consensus information is absorbed by electronic markets almost instantly, leaving virtually zero spread in large caps ($+0.21\%$, $t = 0.103$, $p = 0.9178$). Competing on raw parsing speed produces zero net profit after compute costs.
- **2. Embrace Controversy (Disagreement Pays):** Across 960 evaluations over 4 years, stocks with high AI disagreement delivered a **+178.4% cumulative return**, outperforming high-agreement stocks (**+148.7%**) by 29.7 percentage points and beating the S&P 500 (**+101.9%**) by 76.5 percentage points. Disagreement is not noise; it highlights complexity and uncertainty where standard models clash—and the market pays an excess return for bearing that controversy.
- **3. Beware the Algorithmic Herd:** When funds run similar models trained on similar data, they end up holding correlated portfolios. When a macro shock hits, identical stop-losses trigger synchronized selling and flash crashes. Durable edge requires building quantitative models that are intentionally independent ($r \approx 0$) from standard LLM consensus.
- **4. Go Where AI Cannot See:** Real informational edge has moved away from online text to physical ground reality: counting factory gate dispatches, checking dealer networks, forensic promoter governance analysis, and trading against retail liquidity flows (like the ₹20,000+ Cr/month Indian SIP wall). Human judgment grounded in physical reality remains irreplaceable.

Study Scope & Limitations

- **1. Calibrated Model Personas:** Tested with 4 stylized deterministic personas (Momentum, Value, Quality, Macro) to ensure strict mathematical repeatability across 960 evaluations.
- **2. Time Horizon & Sample:** Covers 60 equities across 16 quarters (2022–2026). Future research should expand across the Russell 3000 over multi-decade cycles.
- **3. Fixed Weights:** The Consensus Index uses fixed weights ($w_d = 0.60$, $w_t = 0.40$). Dynamic weighting across volatility regimes is an active research area.
- **4. Trading Costs:** While the Small-Cap spread ($+1.37\%$) survives transaction costs up to 50 bps, the Large-Cap spread ($+0.21\%$) disappears with fees above 18 bps.
- **5. Data Coverage:** 4 small-cap stocks were excluded from dispersion tests due to missing sell-side target prices on Yahoo Finance ($N = 54$).
- **6. Appendices & Citations:** Full mathematical derivations, econometric proofs, and all 22 annotated academic citations are provided in the companion Technical Appendices & Citations document (PDF).

Research Replication & Code Repository: <https://github.com/Parth777777/ai-model-consensus-market-efficiency>

Companion Technical Appendices, Citations PDF & Full Simulation Pipeline Available Online.